

PROJECT DOCUMENTATION

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

8 PHASE PROJECT DOCUMENTATION

ServiceNow – Flow Designer

TEAM DETAILS

S.No	Role	Name	E-Mail ID
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PHASE 1: BRAINSTORMING & IDEATION PHASE

1.1 Problem Identification

The school IT helpdesk receives multiple incident requests every day from students and teachers. These requests may be related to Wi-Fi issues, projector failures, password problems, and slow computers.

Currently, IT staff manually review each request and assign the appropriate category and subcategory. This process is time-consuming and can become inefficient when the number of requests increases.

Therefore, the idea was developed to automate the classification of IT support tickets using ServiceNow Flow Designer. The system analyzes keywords in the ticket's short description and automatically assigns the appropriate category and subcategory.

1.2 Project Idea

The proposed project is Auto Ticket Classification using Flow Designer.

The system automatically identifies the type of IT issue from the description provided by the user and classifies the ticket accordingly.

Keyword / Issue	Category	Subcategory
WiFi / Network	Network	Wi-Fi
Projector	Hardware	Projector
Password / Login	Access	Forgot Password
Slow / Hanging	Performance	Slow Computer

The project uses a no-code approach through ServiceNow Flow Designer, making the solution easier to maintain and extend.

1.3 Objectives

- Automatically classify incidents when they are created.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Assign Category and Subcategory automatically.
- Send an automated email notification to the caller.
- Provide a structured and maintainable solution.

PHASE 2: REQUIREMENT ANALYSIS PHASE

2.1 Existing System

In the existing process, IT staff manually review each support ticket and identify the issue type, select the correct category, select the appropriate subcategory, and inform the caller about ticket creation.

This manual process can be time-consuming, error-prone, and difficult to scale when the number of requests increases.

2.2 Proposed System

The proposed system automates ticket classification using ServiceNow Flow Designer. When a new ticket is created, the system checks the issue description for predefined keywords and automatically assigns the relevant Category and Subcategory.

2.3 Functional Requirements

1. Automatically classify IT tickets based on the issue description.
2. Assign Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller.
5. Store ticket information in a structured format.
6. Support easy maintenance and future scalability.

2.4 Technologies Used

Item	Details
Platform	ServiceNow
Automation Tool	Flow Designer
Approach	No-code automation
Main Function	Automatic IT ticket classification

PHASE 3: PROJECT DESIGN PHASE

3.1 System Design

The project is designed around a custom ticket table called Incident Workflow. The table stores the required ticket information and contains fields such as:

Field	Type
Number	Auto Number
Caller	Reference – sys_user
Category	Choice
Subcategory	Choice
Short Description	String
Description	String
State	Choice
Assigned Group	Reference – sys_user_group
Assigned to	Reference – sys_user

3.2 Category and Subcategory Design

- Network
- Hardware
- Access
- Performance

The corresponding subcategories are:

- Wi-Fi → Network
- Projector → Hardware
- Forgot Password → Access
- Slow Computer → Performance

The Subcategory field is made dependent on the Category field, so the available subcategory options can be associated with the selected category.

3.3 Automation Design

The basic automation flow is:

Ticket Created → Flow Triggered → Short Description Checked → Keyword Identified → Category/Subcategory Updated → Email Notification Sent

This design allows the classification process to happen automatically without requiring the user to manually select the category and subcategory.

PHASE 4: PROJECT PLANNING PHASE

4.1 Project Activities

7. Identify the IT helpdesk problem.
8. Analyze business requirements.
9. Create a ServiceNow Update Set.
10. Create the custom ticket table.
11. Create and configure required fields.
12. Implement Category–Subcategory dependency.
13. Create the Flow Designer automation.
14. Configure keyword-based classification.
15. Configure email notification.
16. Test the complete workflow.
17. Complete the Update Set.
18. Export the Update Set for deployment/sharing.

4.2 Update Set Planning

Property	Value
Update Set Name	Project Update Set
Initial State	In progress
Application	Global

PHASE 5: PROJECT DEVELOPMENT PHASE

5.1 Creation of Custom Table

A custom table named Incident Workflow is created to store ticket records. The table is configured with an auto-number field for ticket identification.

5.2 Field Creation

The required fields are created with appropriate data types and choices. The Category, Subcategory, and State fields are configured as choice fields, while Caller, Assigned Group, and Assigned to use reference fields.

5.3 Dependent Fields

The Subcategory field is configured as a dependent field based on Category.

Category → Subcategory

Category	Subcategory
Network	Wi-Fi
Hardware	Projector

Access	Forgot Password
Performance	Slow Computer

5.4 Flow Designer Creation

A new Flow Designer flow is created with the name Auto Classify School IT Tickets. The application is set to Global.

5.5 Trigger Configuration

Setting	Value
Trigger	Record Created
Table	Incident Workflow
Condition	Category is Empty

5.6 Wi-Fi Classification

If the Short Description contains Wi-Fi or Network:

- Category → Network
- Subcategory → Wi-Fi

5.7 Projector Classification

If the Short Description contains Projector:

- Category → Hardware
- Subcategory → Projector

5.8 Password Classification

If the Short Description contains Forgot Password:

- Category → Access
- Subcategory → Forgot Password

5.9 Slow Computer Classification

If the Short Description contains Slow Computer:

- Category → Performance
- Subcategory → Slow Computer

5.10 Email Notification

Setting	Value
Action	Send Email
Recipient	Caller → Email
Subject	Your Request for the issue has been submitted.
Body	Ticket confirmation message.

After completing the flow configuration, the flow is activated.

PHASE 6: PROJECT TESTING PHASE

6.1 Testing Objective

Testing is performed to ensure that the developed solution works as expected, satisfies the functional requirements, validates ticket data, and maintains appropriate access and security practices.

6.2 Data Validation

- Mandatory fields are captured correctly.
- Auto-number is generated without duplication.
- Category and Subcategory are stored accurately.
- Reference fields resolve correctly to users and groups.

6.3 Test Scenario 1 – Wi-Fi Issue

Test Item	Value
Caller	Test User
Short Description	WiFi not working in library
Expected Category	Network
Expected Subcategory	WiFi
Expected Notification	Email sent to caller

6.4 Email Notification Testing

The email notification is verified by opening All → Emails → System Logs → Emails. The email record can then be opened and the Preview Mail option can be used to check the notification.

6.5 Test Scenario 2 – Projector Issue

Test Item	Value
Caller	Test User
Short Description	Projector not turning on
Expected Category	Hardware
Expected Subcategory	Projector
Expected Notification	Email sent to caller

PHASE 7: PROJECT DOCUMENTATION PHASE

7.1 Documentation Preparation

The project documentation contains the problem statement, objectives, requirements, system design, development activities, Flow Designer configuration, testing results, deployment procedure, and conclusion.

The documentation provides a reference for understanding how the automated ticket classification system was created and configured.

7.2 Update Set Completion

After development and testing, the Project Update Set is opened from Local Update Sets. The state is changed from In Progress to Complete.

The completed Update Set can then be exported using the Export to XML option.

7.3 Project Documentation Contents

- Team details
- Project title
- Problem statement
- Objectives
- Requirement analysis
- Project design
- Project planning
- Development steps
- Flow Designer configuration
- Testing scenarios
- Testing results
- Deployment/Update Set information
- Screenshots
- Conclusion

PHASE 8: PROJECT DEMONSTRATION PHASE

8.1 Demonstration Objective

The project demonstration shows the complete working flow of the Auto Ticket Classification using Flow Designer system.

8.2 Demonstration Flow

19. Open the Incident Workflow ticket form.
20. Enter the Caller name.
21. Enter an issue in Short Description.
22. Save the ticket.
23. Show that the system automatically assigns Category and Subcategory.
24. Show the email notification sent to the caller.

8.3 Demonstration – Wi-Fi Issue

Example Short Description: WiFi not working in library

- Category → Network
- Subcategory → Wi-Fi
- Email → Sent to caller's email ID

8.4 Demonstration – Projector Issue

Example Short Description: Projector not turning on

- Category → Hardware
- Subcategory → Projector
- Email → Sent to caller's email ID

CONCLUSION

The Auto Ticket Classification using Flow Designer project provides an end-to-end automation solution for a school IT helpdesk. The system automatically classifies IT tickets based on the issue description and reduces the manual effort required for ticket categorization.

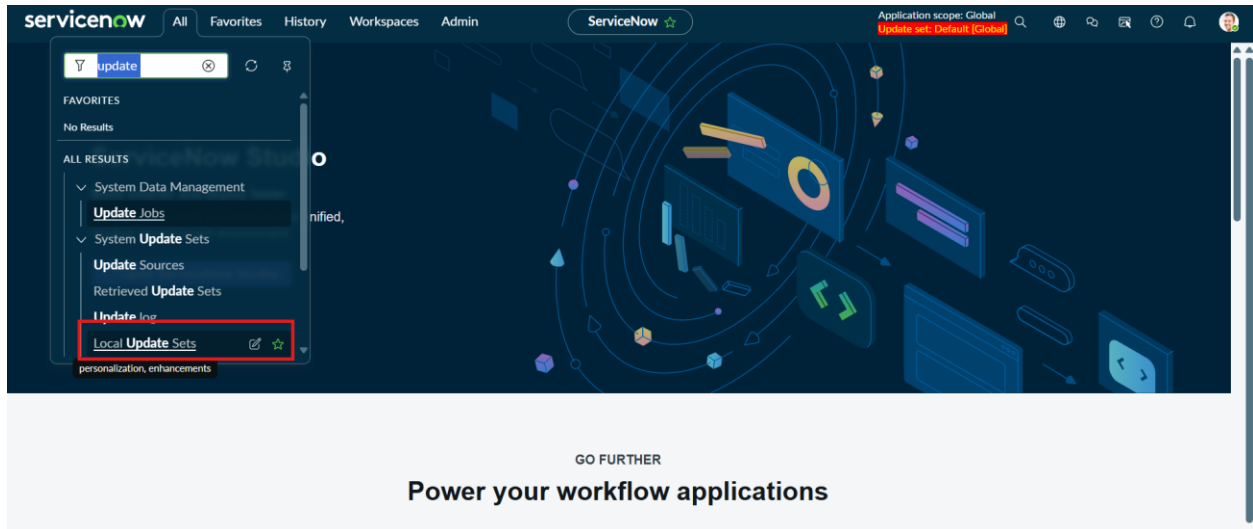
The project uses dependent choice fields to maintain structured ticket information and Flow Designer to automate classification. Automated email notifications also provide confirmation to the caller after ticket creation.

The solution is implemented using a no-code approach and is designed to be maintainable and scalable. The project can be extended in the future with features such as assignment automation, SLA tracking, or predictive intelligence.

SCREENSHOT / EVIDENCE SECTION

Insert the actual ServiceNow screenshots below the relevant Phase/Activity headings in the final submission.

Phase 1 – Ideation / Problem Identification Screenshot



Phase 2 – Requirement / Business Use Case Screenshot

The screenshot shows the 'Update Set' form in ServiceNow. The form has a header bar with a back arrow, a menu icon, and the text 'Update Set New record'. On the right side of the header bar, there are icons for edit, info, and a list of actions including 'Submit' and 'Submit and Make Current'. The form fields are as follows: 'Name' is a text field with the value 'Project Update Set' and a red box around it; 'State' is a dropdown menu with the value 'In progress'; 'Parent' is a text field with a search icon; 'Release date' is a date picker; and 'Description' is a large text area. At the bottom of the form, there are two buttons: 'Submit' and 'Submit and Make Current'. The background is a light gray.

Update Set
New record

* Name

State

Parent

Release date

Description

Application

Phase 3 – Table and Field Design Screenshots

servicenow

All Favorites History Workspaces Admin

ServiceNow

Application scope: Global
Update set: Project Update Set [Global]

tables

- Archive Audit Result
- Archive Knowledge Use
- System Clone
- Clone Definition
- Exclude Tables
- System Definition
 - Tables**
 - Tables & Columns
 - Table Rotations
 - Decision Tables

GO FURTHER

Power your workflow applications

Table
New record

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Incident Workflow

* Name u_incident_workflow

Extends table

Application Global

Create module

Remote Table

Columns

Controls

Application Access

Table Columns

for text

Search

Dictionary Entries

Column label

Type

Reference

Max length

Default value

Display

Insert a new row...

Submit

Cancel

Related Links

Label	Name	Extends table	Extensible	Updated
Search	Search	Search	Search	Search
Logged in User	v_user_session	(empty)	false	2007-07-13 16:15:23
System Plugin	v_plugin	(empty)	false	2007-10-09 12:10:05
Active Transactions	v_transaction	(empty)	false	2008-05-23 17:30:17
Node Log	v_logfiles	(empty)	false	2009-05-14 17:05:41
Customer Uploads	v_customer_uploads	(empty)	false	2009-05-14 18:38:13
MySQL Process List	v_mysql_proclist	(empty)	false	2012-02-25 19:47:02
MySQL Status	v_mysql_status	(empty)	false	2012-02-25 19:47:08
MySQL Variables	v_mysql_variables	(empty)	false	2012-02-25 19:47:12
IOStats	v_iostats	(empty)	false	2012-02-27 17:13:51
Cluster Nodes	v_cluster_nodes	(empty)	false	2012-02-27 23:48:26
SQL Debug	v_sql_debug	(empty)	false	2012-02-28 16:40:55
Workflow Validation Report	v_wf_validation_report	(empty)	false	2012-10-01 10:43:05
Hyper-V Resource Pool Component	cmdb_ci_hyper_v_rpool_comp	Hyper-V Object	false	2012-10-30 08:07:25
Application Menu	sys_app_application	Application File	false	2012-10-30 13:59:18
Menu Category	sys_app_category	Application File	false	2012-10-30 13:59:44
Scriptable Object	v_scriptable_object	(empty)	false	2012-11-09 15:15:23

1 to 20 of 5,827

Table
Incident Workflow

Delete Update Delete All Records

Columns

Controls

Application Access

Extensible

Live feed

Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format

Auto-number

Prefix INC

Number 500

Number of digits 5

Security Rules (ACLs) are required if anyone other than an administrator needs to work with this table. Creating default security rules will grant full access to this table to anyone with the user role you specify.

Create access controls

* User role u_incident_workflow_user

Delete

Update

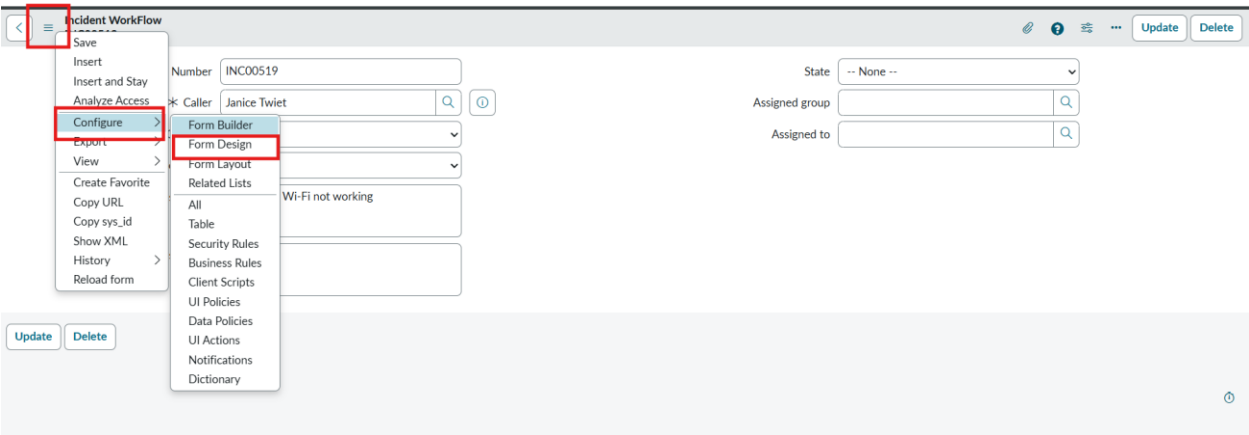
Delete All Records

Related Links

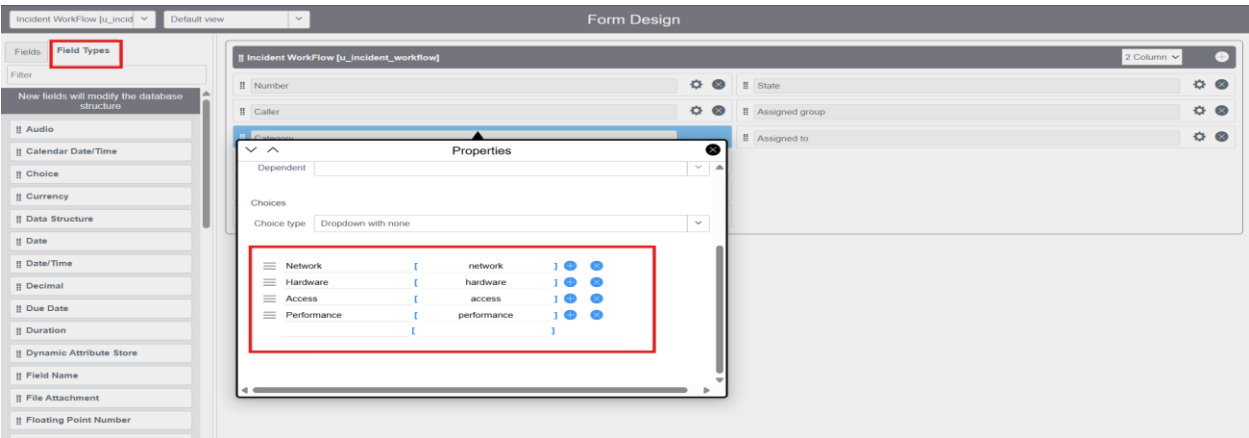
Form Builder

Default Form

Phase 4 – Planning / Update Set Screenshot



Phase 5 – Flow Designer Development Screenshots



Phase 6 – Testing Result Screenshots

Incident Workflow [u_incident_workflow] Default view Form Design

Fields Field Types

Filter

New fields will modify the database structure

|| Audio

|| Calendar Date/Time

|| Choice

|| Currency

|| Data Structure

|| Date

|| Date/Time

|| Decimal

|| Due Date

|| Duration

|| Dynamic Attribute Store

|| Field Name

|| File Attachment

|| Floating Point Number

|| Incident Workflow [u_incident_workflow]

|| Number

|| Caller

|| Category

|| State

|| Assigned group

|| Assigned to

Properties

Dependent

Choices

Choice type: Dropdown with none

Wi-Fi	wi-fi		
Projector	projector		
Forgot Password	forgot password		
Slow Computer	slow computer		

Phase 7 – Completed Update Set / Documentation Screenshot

Update Set Project Update Set

* Name Project Update Set

State In progress

Parent In progress

Release date Ignore

Install date

Installed from

Description

Application Global

Created 2026-02-06 20:55:31

Created by admin

Merged to

Update

Related Links

Merge With Another Update Set

Scan Update Set

[SN Utils] Versions (0)

Update Set Project Update Set

* Name Project Update Set

State Complete

Parent

Release date

Install date

Installed from

Description

Application Global

Created 2026-02-06 20:55:31

Created by admin

Merged to

Update Back Out

Related Links

Export to XML

Merge With Another Update Set

Scan Update Set

[SN Utils] Versions (0)

Phase 8 – Final Demonstration Screenshots

Browser address bar: d8%26sysparm_view%3D%26sysparm_domain%3Dnull...

Page header: atGPT Candidate Home

Application Update set:

Recent download history

File Name	Size	Status
sys_remote_update_set_a7b40de6fc590ceaad36c.xml	182 KB	Done

Full download history

ication: Global

reated: 2026-02-06 20:55:31

ted by: admin